Q1

```sql
use [24L-0583 Lab 4];

-- Q1
SELECT fname, lname, ssn FROM employee
WHERE fname LIKE 'J%';
```

146 %

Results | Messages

	fname	lname	ssn
1	John	Smith	123456789
2	Jane	Doe	987654321

Q2

24L-0583 Lab 4.sql -...0583 Lab 4 (sa (71))

```sql
use [24L-0583 Lab 4];

-- Q1
SELECT fname, lname, ssn FROM employee
WHERE fname LIKE 'J%';

-- Q2
SELECT fname, lname, ssn FROM employee
WHERE lname LIKE '%son';
```

146 %

Results | Messages

	fname	lname	ssn
1	Sara	Wilson	159753486
2	Sophia	Anderson	753951486

Q3

24L-0583 Lab 4.sql - ...0583 Lab 4 (sa (71))

```sql
use [24L-0583 Lab 4];

-- Q1
SELECT fname, lname, ssn FROM employee
WHERE fname LIKE 'J%';

-- Q2
SELECT fname, lname, ssn FROM employee
WHERE lname LIKE '%son';

-- Q3
SELECT fname, lname, ssn FROM employee
WHERE fname LIKE 'A%' OR fname LIKE 'S%';
```

146 %

Results | Messages

	fname	lname	ssn
1	Sara	Wilson	159753486
2	Sophia	Anderson	753951486

Q4

24L-0583 Lab 4.sql - ...0583 Lab 4 (sa (71))

```sql
-- Q2
SELECT fname, lname, ssn FROM employee
WHERE lname LIKE '%son';

-- Q3
SELECT fname, lname, ssn FROM employee
WHERE fname LIKE 'A%' OR fname LIKE 'S%';

-- Q4
SELECT TOP 5 * FROM employee;
```

146 %

Results | Messages

	fname	minit	lname	ssn	bdate	address	sex	salary	supersan	dno
1	Tom	K	Holland	112233445	1993-11-11	789 Cedar St	M	57000.00	456123789	2
2	John	D	Smith	123456789	1980-01-01	123 Main St	M	50000.00	NULL	1
3	Sara	F	Wilson	159753486	1992-06-06	987 Spruce St	F	47000.00	456123789	6
4	Lucy	L	Miller	223344556	1996-12-12	999 Birch St	F	60000.00	987654321	5
5	Chris	E	Davis	321654987	1988-05-05	654 Maple St	M	65000.00	987654321	5

Q5

24L-0583 Lab 4.sql - ...0583 Lab 4 (sa (71))

```sql
-- Q2
SELECT fname, lname, ssn FROM employee
WHERE lname LIKE '%son';

-- Q3
SELECT fname, lname, ssn FROM employee
WHERE fname LIKE 'A%' OR fname LIKE 'S%';

-- Q4
SELECT TOP 5 * FROM employee;

-- Q5
SELECT employee.fname, employee.lname, department.dname
FROM employee
FULL OUTER JOIN department
ON employee.dno = department.dnumber;
```

146 %

Results | Messages

	fname	lname	dname
1	Tom	Holland	Finance
2	John	Smith	HR
3	Sara	Wilson	R&D
4	Lucy	Miller	Operations
5	Chris	Davis	Operations
6	Emily	Jones	IT
7	Olivia	Moore	Logistics
8	Sophia	Anderson	Legal
9	Michael	Brown	Marketing
10	Ethan	Thomas	Support
11	David	Taylor	Sales
12	Jane	Doe	Finance
13	NULL	NULL	Public Relations
14	NULL	NULL	Strategy

Q6

24L-0583 Lab 4.sql - ...0583 Lab 4 (sa (71))

```sql
SELECT TOP 5 * FROM employee;

-- Q5
SELECT employee.fname, employee.lname, department.dname
FROM employee
FULL OUTER JOIN department
ON employee.dno = department.dnumber;

-- Q6
SELECT employee.fname, employee.lname, COUNT(works_on.pno) AS project_count
FROM employee
JOIN works_on ON employee.ssn = works_on.essn
GROUP BY employee.fname, employee.lname
HAVING COUNT(works_on.pno) > 1;
```

146 %

Results | Messages

	fname	lname	project_count
1	Jane	Doe	2
2	John	Smith	2
3	David	Taylor	2

Q7

24L-0583 Lab 4.sql - ...0583 Lab 4 (sa (71))

```sql
FULL OUTER JOIN department
ON employee.dno = department.dnumber;

-- Q6
SELECT employee.fname, employee.lname, COUNT(works_on.pno) AS project_count
FROM employee
JOIN works_on ON employee.ssn = works_on.essn
GROUP BY employee.fname, employee.lname
HAVING COUNT(works_on.pno) > 1;

-- Q7
SELECT employee.fname, employee.lname, dependent.dependent_name
FROM employee
LEFT JOIN dependent ON employee.ssn = dependent.essn;
```

146 %

Results | Messages

	fname	lname	dependent_name
1	Tom	Holland	NULL
2	John	Smith	Anna
3	John	Smith	Mark
4	Sara	Wilson	Emma
5	Lucy	Miller	NULL
6	Chris	Davis	Liam
7	Emily	Jones	Ella
8	Olivia	Moore	NULL
9	Sophia	Anderson	Lucas
10	Michael	Brown	Max
11	Ethan	Thomas	Sophia
12	David	Taylor	Olive
13	Jane	Doe	Sam

Q8

24L-0583 Lab 4.sql - ...0583 Lab 4 (sa (71))

```sql
-- Q7
SELECT employee.fname, employee.lname, dependent.dependent_name
FROM employee
LEFT JOIN dependent ON employee.ssn = dependent.essn;

-- Q8
SELECT department.dname, AVG(employee.salary) AS avg_salary
FROM department
JOIN employee ON department.dnumber = employee.dno
GROUP BY department.dname
ORDER BY avg_salary DESC;
```

146 %

Results | Messages

	dname	avg_salary
1	Marketing	75000.000000
2	Operations	62500.000000
3	Logistics	62000.000000
4	Finance	58500.000000
5	Support	56000.000000
6	IT	55000.000000
7	Sales	52000.000000
8	HR	50000.000000
9	Legal	48000.000000
10	R&D	47000.000000

Q9

```sql
-- Q7
SELECT employee.fname, employee.lname, dependent.dependent_name
FROM employee
LEFT JOIN dependent ON employee.ssn = dependent.essn;

-- Q8
SELECT department.dname, AVG(employee.salary) AS avg_salary
FROM department
JOIN employee ON department.dnumber = employee.dno
GROUP BY department.dname
ORDER BY avg_salary DESC;

-- Q9
SELECT fname, lname, salary FROM employee
ORDER BY salary ASC;
```

Results | Messages

	fname	lname	salary
1	Sara	Wilson	47000.00
2	Sophia	Anderson	48000.00
3	John	Smith	50000.00
4	David	Taylor	52000.00
5	Emily	Jones	55000.00
6	Ethan	Thomas	56000.00
7	Tom	Holland	57000.00
8	Lucy	Miller	60000.00
9	Jane	Doe	60000.00
10	Olivia	Moore	62000.00
11	Chris	Davis	65000.00
12	Michael	Brown	75000.00

Q10

```sql
GROUP BY department.dname
ORDER BY avg_salary DESC;

-- Q9
SELECT fname, lname, salary FROM employee
ORDER BY salary ASC;

-- Q10
SELECT dname FROM department
WHERE dnumber NOT IN
(SELECT dnum FROM project);
```

Results | Messages

	dname
1	Public Relations
2	Strategy

Q11

```sql
-- Q9
SELECT fname, lname, salary FROM employee
ORDER BY salary ASC;

-- Q10
SELECT dname FROM department
WHERE dnumber NOT IN
(SELECT dnum FROM project);

-- Q11
SELECT department.dname, COUNT(employee.ssn) AS total_employees
FROM department
LEFT JOIN employee ON department.dnumber = employee.dno
GROUP BY department.dname;
```

Results | Messages

	dname	total_employees
1	Finance	2
2	HR	1
3	IT	1
4	Legal	1
5	Logistics	1
6	Marketing	1
7	Operations	2
8	Public Relations	0
9	R&D	1
10	Sales	1
11	Strategy	0
12	Support	1

Q12

```sql
-- Q11
SELECT department.dname, COUNT(employee.ssn) AS total_employees
FROM department
LEFT JOIN employee ON department.dnumber = employee.dno
GROUP BY department.dname;

-- Q12
SELECT department.dname, department.dnumber, employee.fname, employee.lname, employee.ssn
FROM department
LEFT JOIN employee ON department.dnumber = employee.dno;
```

Results | Messages

	dname	dnumber	fname	lname	ssn
1	HR	1	John	Smith	123456789
2	Finance	2	Tom	Holland	112233445
3	Finance	2	Jane	Doe	987654321
4	IT	3	Emily	Jones	456123789
5	Marketing	4	Michael	Brown	789456123
6	Operations	5	Lucy	Miller	223344556
7	Operations	5	Chris	Davis	321654987
8	R&D	6	Sara	Wilson	159753486
9	Sales	7	David	Taylor	951357486
10	Legal	8	Sophia	Anderson	753951486
11	Support	9	Ethan	Thomas	852741963
12	Logistics	10	Olivia	Moore	654852963
13	Public Relations	11	NULL	NULL	NULL
14	Strategy	12	NULL	NULL	NULL

Q13

```sql
GROUP BY department.dname;

-- Q12
SELECT department.dname, department.dnumber, employee.fname, employee.lname, employee.ssn
FROM department
LEFT JOIN employee ON department.dnumber = employee.dno;

-- Q13
SELECT project.pname, project.pnumber, employee.fname, employee.lname, employee.ssn, works_on.hours
FROM project
LEFT JOIN works_on ON project.pnumber = works_on.pno
LEFT JOIN employee ON works_on.essn = employee.ssn;
```

	pname	pnumber	fname	lname	ssn	hours
1	Alpha	1	John	Smith	123456789	20.5
2	Beta	2	John	Smith	123456789	10.0
3	Beta	2	Jane	Doe	987654321	30.0
4	Gamma	3	Emily	Jones	456123789	25.0
5	Delta	4	Michael	Brown	789456123	35.0
6	Delta	4	Jane	Doe	987654321	15.5
7	Epsilon	5	Chris	Davis	321654987	40.0
8	Zeta	6	Sara	Wilson	159753486	15.0
9	Zeta	6	David	Taylor	951357486	10.0
10	Eta	7	David	Taylor	951357486	22.0
11	Theta	8	Sophia	Anderson	753951486	18.0
12	Iota	9	Ethan	Thomas	852741963	28.5
13	Kappa	10	Olivia	Moore	654852963	33.0
14	Lambda	11	NULL	NULL	NULL	NULL
15	Omega	12	NULL	NULL	NULL	NULL

Q14

```sql
-- Q12
SELECT department.dname, department.dnumber, employee.fname, employee.lname, employee.ssn
FROM department
LEFT JOIN employee ON department.dnumber = employee.dno;

-- Q13
SELECT project.pname, project.pnumber, employee.fname, employee.lname, employee.ssn, works_on.hours
FROM project
LEFT JOIN works_on ON project.pnumber = works_on.pno
LEFT JOIN employee ON works_on.essn = employee.ssn;

-- Q14
SELECT department.dname, department.dnumber, employee.fname, employee.lname, employee.ssn
FROM department
LEFT JOIN employee ON department.mgrssn = employee.ssn;
```

	dname	dnumber	fname	lname	ssn
1	HR	1	John	Smith	123456789
2	Finance	2	Jane	Doe	987654321
3	IT	3	Emily	Jones	456123789
4	Marketing	4	Michael	Brown	789456123
5	Operations	5	Chris	Davis	321654987
6	R&D	6	Sara	Wilson	159753486
7	Sales	7	David	Taylor	951357486
8	Legal	8	Sophia	Anderson	753951486
9	Support	9	Ethan	Thomas	852741963
10	Logistics	10	Olivia	Moore	654852963
11	Public Relations	11	NULL	NULL	NULL
12	Strategy	12	NULL	NULL	NULL

Q15

(No record satisfies the condition)

```sql
-- Q14
SELECT department.dname, department.dnumber, employee.fname, employee.lname, employee.ssn
FROM department
LEFT JOIN employee ON department.mgrssn = employee.ssn;

-- Q15
SELECT department.dname, COUNT(employee.ssn) AS employee_count
FROM department
JOIN employee ON department.dnumber = employee.dno
GROUP BY department.dname
HAVING COUNT(employee.ssn) > 3;
```

Results | Messages

dname	employee_count

Q16

```sql
LEFT JOIN employee ON department.mgrssn = employee.ssn;

-- Q15
SELECT department.dname, COUNT(employee.ssn) AS employee_count
FROM department
JOIN employee ON department.dnumber = employee.dno
GROUP BY department.dname
HAVING COUNT(employee.ssn) > 3;

-- Q16
SELECT dlocation, COUNT(dnumber) AS department_count
FROM dept_locations
GROUP BY dlocation
HAVING COUNT(dnumber) > 1;
```

Results | Messages

	dlocation	department_count
1	Seattle	2

Practice

24L-0583 Lab 4.sql - ...0583 Lab 4 (sa (71))

```sql
SELECT dlocation, COUNT(dnumber) AS department_count
FROM dept_locations
GROUP BY dlocation
HAVING COUNT(dnumber) > 1;

-- Practice
SELECT department.dname, SUM(works_on.hours) AS total_hours
FROM department
JOIN employee ON department.dnumber = employee.dno
JOIN works_on ON employee.ssn = works_on.essn
GROUP BY department.dname
HAVING SUM(works_on.hours) >
(SELECT AVG(hours) FROM works_on)
ORDER BY total_hours DESC;
```

146 %

Results | Messages

	dname	total_hours
1	Finance	45.5
2	Operations	40.0
3	Marketing	35.0
4	Logistics	33.0
5	Sales	32.0
6	HR	30.5
7	Support	28.5
8	IT	25.0